

Advanced (Habanero)

- Q1.** What is the GCF of 12 and 18?
(A) 1
(B) 2
(C) 3
(D) 6
(E) 12
- Q2.** What is the LCM of 9 and 12?
(A) 9
(B) 12
(C) 18
(D) 27
(E) 36
- Q3.** What is the GCF of 24 and 30 using prime factorization?
(A) 1
(B) 2
(C) 3
(D) 6
(E) 12
- Q4.** A bookshelf has 12 shelves, and each shelf can hold 8 books. What is the LCM of 12 and 8?
(A) 12
(B) 16
(C) 24
(D) 32
(E) 48
- Q5.** What is the GCF of 15 and 25 using factors?
- (A) 1
(B) 5
(C) 10
(D) 15
(E) 25
- Q6.** A group of friends want to share some candy equally. If they have 18 pieces of candy and there are 3 friends, what is the LCM of 18 and 3?
(A) 3
(B) 6
(C) 9
(D) 12
(E) 18
- Q7.** What is the GCF of 48 and 60 using prime factorization?
(A) 1
(B) 2
(C) 3
(D) 12
(E) 24
- Q8.** A bakery is making a special batch of cookies. They need to package 12 cookies per box and 8 boxes per case. What is the LCM of 12 and 8?
(A) 12
(B) 16
(C) 24
(D) 32
(E) 48

Open Ended Questions (FRQ)

- Q9.** Tom has 18 boxes of crayons and 12 boxes of markers to pack into carts that hold 6 boxes each. What is the GCF and LCM of 18 and 12, and how many carts will Tom need to pack all the boxes?
- Q10.** A teacher is preparing for a class and needs to package 24 pencils and 30 pens into bags that hold x items each. If the teacher wants to use the least number of bags possible, what is the LCM of 24 and 30, and how many bags will the teacher need?

Chef's Tips

1

- Tip 1: Make sure students understand the concept of GCF and LCM before attempting the test.
- Tip 2: Encourage students to use prime factorization to find the GCF and LCM.
- Tip 3: Remind students to read the questions carefully and choose the correct answer.